

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyse DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text (High)
Assessment Tool Weightage: 30%

Dr

QUESTIONS			Total Marks: 50
Qango	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an	BL1 – Remember	5

	Entity Relationship diagram that captures this information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5
4	Hospital Management System	BL2 – Understand	5

	Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)		
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. □ A department controls a number of PROJECT. Projects have a unique name, number and a single location. □ Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. □ Employee's DEPENDENT are tracked for health insurance purposes (dependantname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5

8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

I __ (K.Nikhil Lakshmi sai / 192424366) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer

Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Answers

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Course : CSA0501

1.UPS Product Tracking Information System

Entities

Shipped_Item

- Item_No (PK)
- Weight
- Dimensions
- Insurance_Amount
- Destination
- Delivery_Date

Retail_Center

- Center_ID (PK)
- Type
- Address

Transportation_Event

- Schedule_No (PK)
- Type
- Delivery_Route

Relationships

- Retail_Center receives Shipped_Item (1 : M)
- Shipped_Item uses Transportation_Event (M : N)

2. Course Management System

Entities

- Course
- Name (PK)
- Department
- C_Number

Section (Weak Entity)

- S_Number (Partial Key)
- Enrollment

Room

- R_Number (PK)
- Capacity
- Building

Exam

- Time

Relationships

- Course has Section (1 : M)
- Section is conducted in Room
- Section has Exam

3. University Registrar's Office

Entities

- Course
- Course_No (PK)
- Title
- Credits

Course_Offering

- Year
- Semester

- Section_No
- Classroom

Student

- Student_ID (PK)
- Name
- Program
- Instructor
- Instructor_ID (PK)
- Name
- Department

Relationships

- Course has Course_Offering (1 : M)
- Instructor teaches Course_Offering (1 : M)
- Student enrolls in Course_Offering (M : N)
- Attribute: Grade

4. Hospital Management System

Entities

Patient

- SS# (PK)
- Name
- Insurance

Physician

- Physician_Name (PK)
- Specialization

Test_Log

- Test_Name
- Date

- Time

Patient_History

- Test_Name
- Date
- Relationship
- Physician treats Patient (M : N)

Tables

- Patient(SS#, Name, Insurance)
- Physician(Physician_Name, Specialization)
- Test_Log(SS#, Test_Name, Date, Time)
- Doctor_Patient(Physician_Name, SS#)
- Patient_History(SS#, Test_Name, Date)

5. Company Project Management System

Entities

- Department
- Employee
- Project
- Dependent
- Relationships
- Department controls Project (1 : M)
- Employee works for Department (M : 1)
- Employee works on Project (M : N)
- Attribute: Hours
- Employee supervises Employee (1 : M)
- Employee has Dependent (1 : M)

6. Schema and Instance

Schema

A Schema is the overall structure or design of a database.

Example: Student(ID, Name, Age)

Instance

An Instance is the actual data stored in the database at a particular time.

Types of Schema

Logical Schema: Defines the database structure (tables, attributes, relationships).

Physical Schema: Describes how data is stored physically.

View Schema: Shows a customized view of the database for different users.

7. Bank Management System

Entities

- Customer
- Account
- Loan
- Branch
- Employee
- Relationships
- Customer owns Account (M : N)
- Customer takes Loan (M : N)
- Branch maintains Account (1 : M)
- Employee works at Branch (M : 1)

8. Extended ER (EER) Model

Definition

The Extended Entity-Relationship (EER) Model is an advanced version of the ER model that supports more complex database designs.

Features

Generalization: Combines similar lower-level entities into a higher-level entity.

Specialization: Divides a higher-level entity into specialized sub-entities.

Aggregation: Treats a relationship as a higher-level entity for further relationships.

9. Hospital ER Diagram

Superclass

- Person
- Subclasses
- Patient
- Doctor
- Weak Entity
- Medical_Record
- Relationships
- Patient owns Medical_Record (1 : M)
- Doctor treats Patient (M : N)

10. Role of Database Administrator (DBA)

Definition

A Database Administrator (DBA) is responsible for managing, maintaining, and securing the database system.

Responsibilities

- Database installation and configuration
- User account and access management
- Database security
- Backup and recovery
- Performance monitoring and tuning
- Maintaining data integrity
- Troubleshooting database issues